

Joris van Oers

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PERSONAL PROFILE

Detail-oriented Biomedical Engineering student with a strong foundation in mechanical design and bioelectronics. Experienced in SolidWorks (3D modeling and prototyping), MATLAB, LTspice, and Arduino-based embedded systems, complemented by a foundational understanding of electronics. Currently completing a BSc in Biomedical Engineering, with an independent project developing a custom-built eye-tracking system featuring self-designed glasses integrated into a motorized cart and combining mechanical design, circuit development, and real-time data capture. Particularly passionate about signal-to-movement engineering: the translation of biological signals into meaningful motion and control, bridging the gap between the nervous system and physical actuation. Eager to deepen this expertise through a Master's in Biomedical Engineering at TU Delft (Neuromuscular track), with the long-term goal of contributing to breakthroughs in neurotechnology and neural-driven assistive systems.

EDUCATION

University of Groningen, Netherlands

Sep 2024 - now

BSc Biomedical Engineering GPA: 7.9

- Minor (ETH Zurich): Robotics & Neuroscience.
- Thesis: Motorized car control with eye-tracking system for people with quadriplegia.

EXPERIENCE

Aeroworks Europe BV - Aerospace & Aviation Technologies, Netherlands Summers 2021, 2022, 2025

Storage Assistant / Manufacturing & Packaging Support

- Designed and implemented a barcode-based logistics scanning system to keep track of jigs and (vacuum) molds across manufacturing stages, making it possible to trace back steps and reducing manual errors via automated tool registration scanning.
- Managed storage operations including inventory organization and incoming and outgoing goods control.
- Performed quality control inspections on manufactured parts, verifying dimensional accuracy and conformity with SolidWorks engineering drawings.

Zorggroep Samen (Elderly care home), Netherlands

Jan 2023 – August 2024

Cleaner

- Taking care of elderly homes, doing heavy tasks that seniors cannot do anymore and making their day with smalltalk.

PROJECTS

Medical device design, CybaNorth Groningen

Redesigning Finger Prosthetic

Nov 2025 – Jan 2026

- Redesigned a finger prosthetic for a myoelectric prosthetic for Team CybaNorth, focusing on improved mechanical performance and wearability.
- Developed and iterated designs using SolidWorks, producing detailed 3D models and engineering drawings to guide the prototyping process.
- Conducted FEM (Finite Element Method) analysis to validate structural integrity and optimize material distribution under simulated load conditions.
- Fabricated and tested physical prototypes, iterating on design improvements based on mechanical testing and functional evaluation results.

Machine Learning Modelling, Groningen

Personal Learning Project

Jun 2025 - Sep 2025

- Built a digit recognition model trained to classify handwritten numbers 1–9, as an introduction to image classification and neural networks.
- Explored core machine learning concepts including data preprocessing, model training, and performance evaluation.
- Gained hands-on experience with ML frameworks and image dataset handling as a foundation for more advanced modelling projects.

SKILLS & LANGUAGES

SolidWorks (CSWA), MATLAB, LTspice, Python, Numpy, Pandas, Latex, Excel, Arduino, basic statistics, Machine Learning (Scikit-learn, XGBoost, Pytorch) | Dutch (native), English C2, Spanish A2, German A2